Kinect Lite: Results

9×9 median (ours) vs. Fu et al. (2012) depth denoising, scored by depth error and camera tracking

ICL-NUIM kt0	synthetic, simulated Kinect noise; frames 1–300; true depth and true camera path
TUM RGB-D fr3 office	real sensor; frames 0–299; true camera path (motion capture), no true depth
trackers	every 2nd frame (150 frames). Depth-only ICP: depth only. RGB-D odometry: colour + depth
colour image	not used by either cleaner; passed unchanged to RGB-D odometry, identical for every method
units	MAE, RMSE, RPE in mm; ATE in cm. Lower is better, except "within 1 cm" and "filled"

1. Every cleaner + tracker combination

rank (TUM)	cleaner	tracker	TUM ATE (cm)	ICL ATE (cm)
1	**median 9×9 (ours)**	RGB-D odometry	**2.08**	0.98
2	**paper (Fu et al.)**	RGB-D odometry	2.10	26.07
3	no cleaning	RGB-D odometry	2.14	23.59
4	**paper (Fu et al.)**	depth-only ICP	3.33	**0.95**
5	**median 9×9 (ours)**	depth-only ICP	13.32	14.94
6	no cleaning	depth-only ICP	13.97	17.07
–	*true depth (ICL only)*	depth-only ICP	–	*0.19*
–	*true depth (ICL only)*	RGB-D odometry	–	*0.81*

source: `src/results/tum_compare.txt, src/results/icl_compare.txt`

Table 1: Camera path error for each combination, sorted by the real-sensor (TUM) result.

2. Depth accuracy (ICL, against the true depth)

method	MAE (mm)	RMSE (mm)	median err. (mm)	within 1 cm (%)	coverage (%)	time, 450 frames
no cleaning (gate only)	39.19	187.45	13.33	42.19	99.16	–
median 9×9 (ours)	**18.77**	**55.41**	**12.32**	**44.66**	99.16	<1 min
median 9×9 + Telea fill	19.34	57.67	12.36	44.47	100.00	<1 min
paper (`center=paper`)	37.69	186.00	13.42	33.69	99.49	≈26 min
paper (`center=exclude`)	37.69	186.00	13.41	33.69	99.49	≈26 min
paper (`center=nonlocal`)	37.69	186.01	13.41	33.69	99.49	≈26 min

source: `src/results/icl_compare.txt (section 1)`

Table 2: Mean over 300 ICL frames. The median lowers the MAE on 100 % of frames, the paper on 99 %.

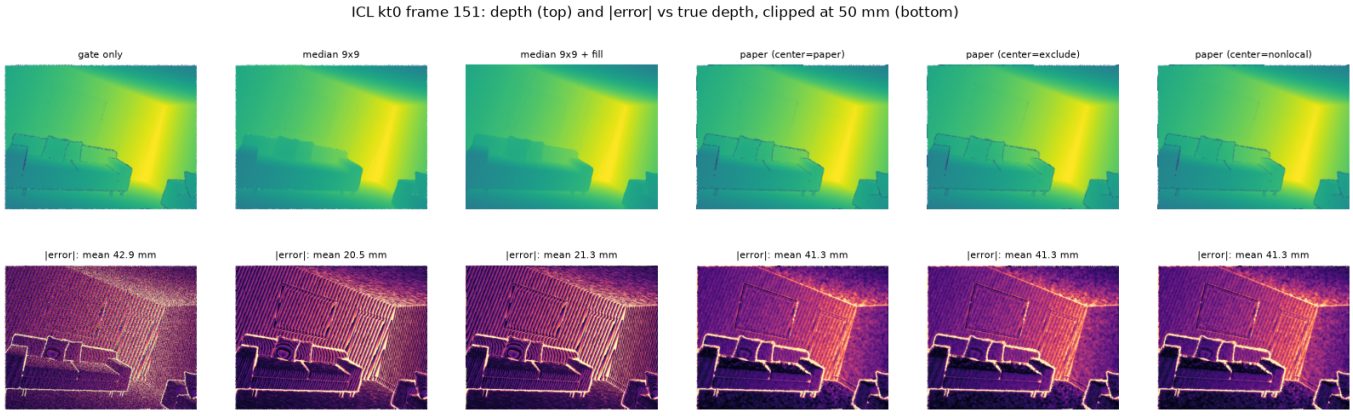


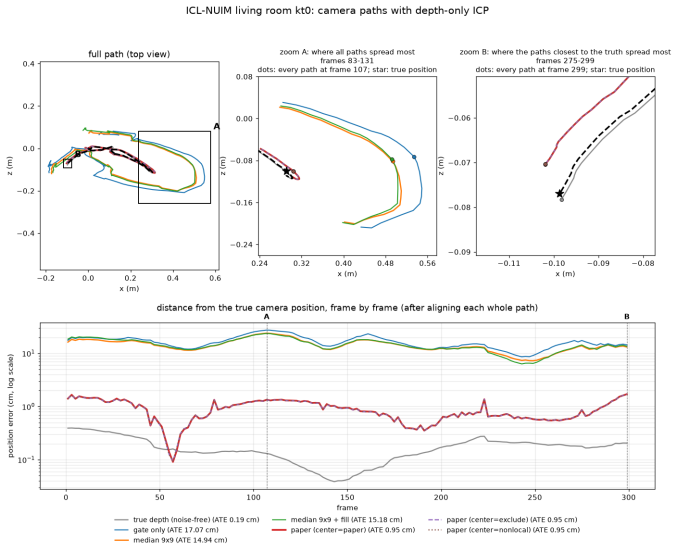
Figure 1: ICL frame 151. Top: depth. Bottom: |error| against the true depth, clipped at 50 mm (brighter = worse).

3. Camera tracking

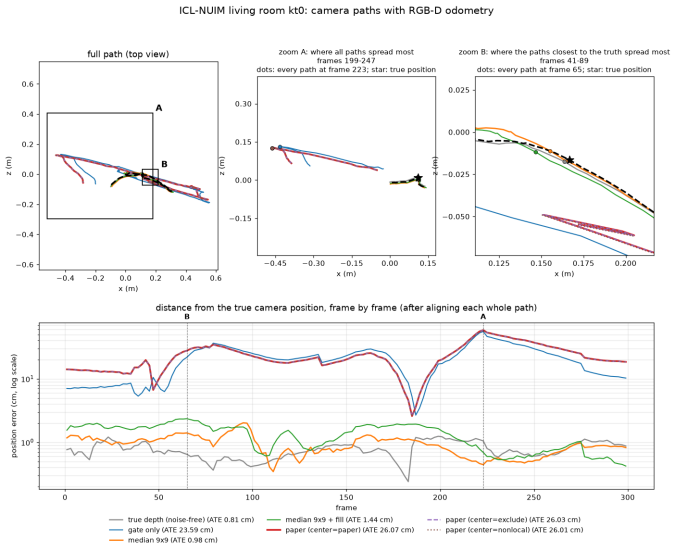
	ICL-NUIM kt0 (synthetic)				TUM fr3 office (real)			
	depth-only ICP		RGB-D odometry		depth-only ICP		RGB-D odometry	
depth used	ATE	RPE	ATE	RPE	ATE	RPE	ATE	RPE
true depth (noise-free)	*0.19*	*0.07*	*0.81*	*2.18*	–	–	–	–
no cleaning (gate only)	17.07	14.43	23.59	24.20	13.97	11.69	2.14	**5.61**
median 9×9 (ours)	14.94	11.78	**0.98**	**1.59**	13.32	10.86	**2.08**	5.69
median 9×9 + Telea fill	15.18	12.01	1.44	1.91	20.20	12.46	3.13	5.69
paper (`center=paper`)	**0.95**	**2.02**	26.07	21.76	3.33	**4.37**	2.10	5.80
paper (`center=exclude`)	**0.95**	**2.02**	26.03	21.76	**3.31**	**4.37**	2.10	5.79
paper (`center=nonlocal`)	**0.95**	**2.02**	26.01	21.75	**3.31**	**4.37**	2.12	5.80

source: `src/results/icl_compare.txt (section 2), src/results/tum_compare.txt (rotational RPE in degrees is also in these files)`

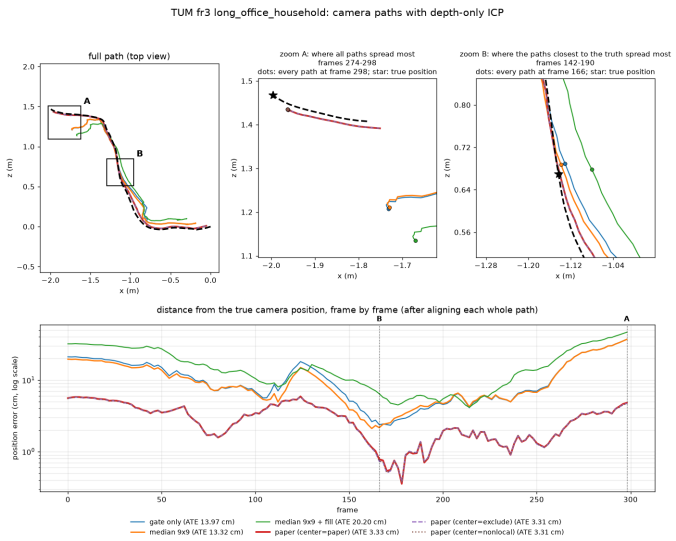
Table 3: 150 tracked frames, ATE in cm, RPE in mm. Bold: best per column, excluding the true-depth row. Camera motion per step: 6.5 mm (ICL, 0.97 m total), 20.7 mm (TUM, 3.09 m total).



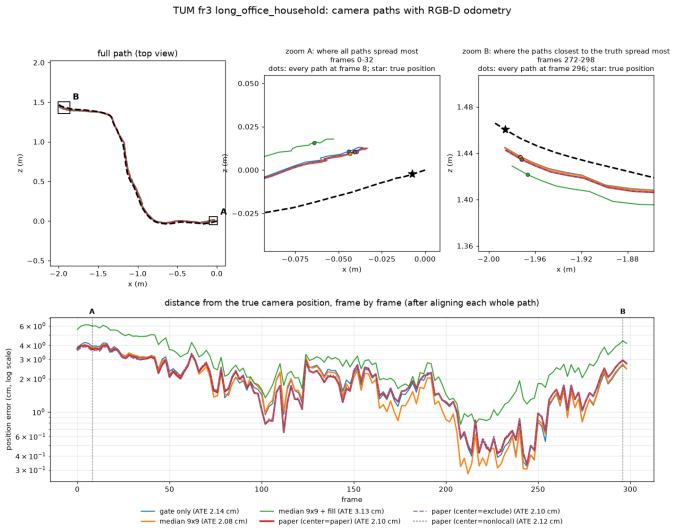
(a) ICL, depth-only ICP



(b) ICL, RGB-D odometry



(c) TUM, depth-only ICP



(d) TUM, RGB-D odometry

Figure 2: Estimated camera paths seen from above with two zoomed regions (top of each panel), and position error per frame on a log scale (bottom of each panel). Legend with each method's ATE under each panel.

4. Hole filling (after the 9×9 median)

	ICL: fill error at real holes			ICL ATE (cm)		TUM: masked test		TUM ATE (cm)		time
fill	MAE (mm)	<5 cm (%)	filled (%)	ICP	RGB-D	MAE (mm)	filled (%)	ICP	RGB-D	(ms/frame)
none (keep holes)	–	–	0.0	**14.94**	**0.98**	–	0.0	13.33	**2.08**	0
Telea	87.0	62.2	100.0	15.18	1.44	625.5	100.0	20.21	3.13	210
Navier–Stokes	22.5	96.8	100.0	15.36	1.46	121.1	100.0	**12.03**	3.89	224
nearest	22.3	97.1	100.0	15.05	1.43	112.3	100.0	17.54	4.16	50
region (small holes)	15.5	95.8	4.0	**14.94**	1.01	11.4	23.2	13.32	2.23	52
region median	22.5	97.3	100.0	15.15	1.43	117.3	100.0	15.14	4.08	386
background	83.1	58.9	100.0	15.24	1.80	3238.8	100.0	13.51	3.14	40
temporal (paper, $t \pm 2$)	**12.2**	**98.8**	88.5	15.11	1.24	**9.1**	39.0	13.37	2.51	7884

source: `experiments/hole_filling/results/fills.txt` (RPE and median errors are also in this file)

Table 4: Holes left after the median: 0.84 % of pixels on ICL, 16.0 % on TUM. TUM has no true depth, so its fill error comes from a masked test (10725 measured pixels per frame hidden and restored). Time measured on TUM.

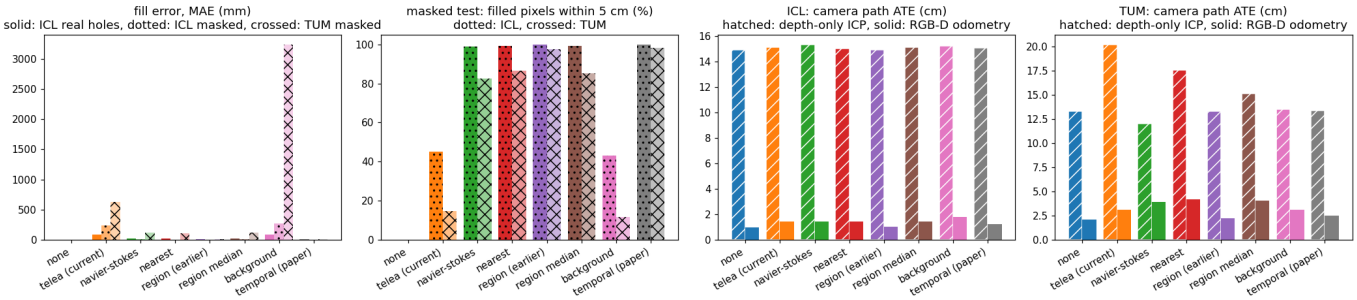


Figure 3: Fill error (left two panels) and camera path ATE (right two panels) for each fill.